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MODULE - 02

FOURIER TRANSFORMS

* Fourier transforms

Fourier transform is a mathematical model which helps to transform the signals between two different domains, such as transforming signal from frequency domain to time domain or vice versa.

→ Fourier transform has many applications in Engineering and physics, such as signal processing, RADAR, and soon.

→ APPLICATIONS

→ Fourier transform plays an important role in different fields such as, voice communication, sonar, earthquake, nuclear science, & even biomedical engineering.

→ It determines the constituent pitches in a musical waveform.

→ It delivers real time pricing while allowing for a realistic structure of asset returns, taking into account excess kurtosis & stochastic volatility.

→ Image analysis, image filtering, image reconstruction & image compression.

* INFINITE FOURIER TRANSFORM (Complex Fourier transform).

The infinite Fourier transform or simply the Fourier transform of a real valued function $f(x)$ is defined by

$$F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{iux} dx \quad \text{--- (1)}$$

provided the integral exists. On integration we obtain a function of u which is usually denoted by $F(u)$ or $\hat{f}(u)$.

* INVERSE FOURIER TRANSFORM

The inverse Fourier transform of $F(u)$ denoted by $F^{-1}[F(u)]$ or $F^{-1}[\hat{f}(u)]$ is defined by the integral

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du \quad \text{--- (2)}$$

On integration we obtain a function of x . that is

$$f(x) = F^{-1}[F(u)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du.$$

* Note :-

① The definitions are deduced from the Fourier integral

$$f(x) = \frac{1}{2\pi} \int_{u=-\infty}^{\infty} \int_{t=-\infty}^{\infty} f(t) e^{iu(t-x)} dt du$$

② In view of the term e^{iux} present in the definition of Fourier transform, it is also called the Complex Fourier transform.

* Properties of Fourier Transform.

① Linearity Property

If $c_1, c_2, \dots, c_n$ are constants then

$$\begin{aligned} F[c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)] \\ = c_1 F[f_1(x)] + c_2 F[f_2(x)] + \dots + c_n F[f_n(x)] \end{aligned}$$

→ By the definition,

$$\begin{aligned} F[c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)] \\ = \int_{-\infty}^{\infty} [c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)] e^{iux} dx \\ = c_1 \int_{-\infty}^{\infty} f_1(x) e^{iux} dx + c_2 \int_{-\infty}^{\infty} f_2(x) e^{iux} dx + \dots + \dots \\ \dots + c_n \int_{-\infty}^{\infty} f_n(x) e^{iux} dx \\ = c_1 F[f_1(x)] + c_2 F[f_2(x)] + \dots + c_n F[f_n(x)] \end{aligned}$$

② Change of Scale Property

If $F[f(x)] = \hat{f}(u)$, then $F[f(ax)] = \frac{1}{a} \hat{f}\left(\frac{u}{a}\right)$

→ By the definition,

$$F[f(ax)] = \int_{-\infty}^{\infty} f(ax) e^{iux} dx$$

put, $ax = t$ $\therefore dx = dt/a$ & t also varies from $-\infty$ to ∞ .

$$\text{now, } F[f(ax)] = \int_{-\infty}^{\infty} f(t) e^{iut/a} \frac{dt}{a}$$

$$\text{i.e., } F[f(ax)] = \frac{1}{a} \int_{-\infty}^{\infty} f(t) e^{i \frac{u}{a} t} dt = \frac{1}{a} \tilde{f}\left(\frac{u}{a}\right) \quad (4)$$

$$\text{Thus, } F[f(ax)] = \frac{1}{a} \tilde{f}\left[\frac{u}{a}\right]$$

(3) Shifting Property

$$\text{If } F[f(x)] = \tilde{f}(u) \text{ then } F[f(x-a)] = e^{iua} \tilde{f}(u)$$

→ By the definition we have,

$$\tilde{f}(u) = F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{iux} dx$$

$$\text{Hence, } F[f(x-a)] = \int_{-\infty}^{\infty} f(x-a) e^{iux} dx.$$

put $x-a=t \therefore dx=dt$, t also varies from $-\infty$ to ∞ .

$$\begin{aligned} \text{now, } F[f(x-a)] &= \int_{-\infty}^{\infty} f(t) e^{iu(t+a)} dt \\ &= e^{iua} \int_{-\infty}^{\infty} f(t) e^{iut} dt = e^{iua} \tilde{f}(u) \end{aligned}$$

$$\text{Thus, } F[f(x-a)] = e^{iua} \tilde{f}(u)$$

(4) Modulation property

$$\text{If } F[f(x)] = \tilde{f}(u) \text{ then } F[f(x) \cos ax] = \frac{1}{2} [\tilde{f}(u+a) + \tilde{f}(u-a)]$$

$$\rightarrow \text{By definition, } F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{iux} dx$$

$$\begin{aligned} \therefore F[f(x) \cos ax] &= \int_{-\infty}^{\infty} f(x) \cos ax e^{iux} dx \\ &= \int_{-\infty}^{\infty} f(x) \cdot \frac{e^{iax} + e^{-iax}}{2} \cdot e^{iux} \cdot dx \end{aligned}$$

$$= \frac{1}{2} \left[\int_{-\infty}^{\infty} f(x) e^{i(u+a)x} dx + \int_{-\infty}^{\infty} f(x) e^{i(u-a)x} dx \right]$$

$$= \frac{1}{2} [\hat{f}(u+a) + \hat{f}(u-a)]$$

Thus, $F[f(x) \cos ax] = \frac{1}{2} [\hat{f}(u+a) + \hat{f}(u-a)]$

* note:- The value of imaginary part $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, $i^6 = -1$

* Problems

① Find the complex Fourier transform of the function.

$$f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases} \quad \text{Hence evaluate } \int_0^{\infty} \frac{\sin x}{x} dx.$$

→ complex Fourier transform of $f(x)$ is given by.

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx.$$

$$F(u) = \int_{-a}^a 1 \cdot e^{iux} dx.$$

$$\text{some } f(x) = \begin{cases} 1 & \text{for } -a \leq x \leq a \\ 0 & \text{otherwise.} \end{cases}$$

$$F(u) = \left[\frac{e^{iux}}{iu} \right]_{x=-a}^a = \frac{1}{iu} \{ e^{iua} - e^{-iua} \}$$

$$F(u) = \frac{1}{iu} \{ (\cos au + i \sin au) - (\cos au - i \sin au) \}$$

$$= \frac{1}{iu} (2i \sin au) = \frac{2 \sin au}{u}$$

Thus, $\boxed{F(u) = \frac{2 \sin au}{u}}$

Let us evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$.

we have obtained, $F(u) = \frac{2 \sin au}{u}$

Inverse Fourier transform is $\frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du = f(x)$.

$$\text{i.e., } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin au}{u} e^{-iux} du = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin au}{u} e^{-iux} du.$$

now, let us put $x=0$.

Since $x=0$ is a point of continuity of $f(x)$, the value of $f(x)$ at $x=0$ being.

$$f(0) = 1 \text{ because } f(x) = 1 \text{ for } |x| \leq a.$$

$$\text{Hence, } \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin au}{u} du = 1, \text{ since } e^0 = 1$$

$$\text{i.e., } \frac{2}{\pi} \int_0^{\infty} \frac{\sin au}{u} du = 1, \text{ since } \frac{\sin au}{u} \text{ is an even function}$$

$$\therefore \int_0^{\infty} \frac{\sin au}{u} du = \frac{\pi}{2}$$

$$\text{Putting } a=1, \int_0^{\infty} \frac{\sin u}{u} du = \frac{\pi}{2}$$

Thus, by changing u to x , we have,

$$\boxed{\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}} //$$

② Find the complex Fourier transform of the function.

$$f(x) = \begin{cases} x, & |x| \leq \alpha \\ 0, & |x| > \alpha \end{cases} \quad \text{where } \alpha \text{ is a positive constant.}$$

$$\rightarrow F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx \quad \text{by data } f(x) = x \text{ for } |x| \leq \alpha.$$

$$= \int_{-\alpha}^{\alpha} x \cdot e^{iux} dx$$

$$= \left[x \cdot \frac{e^{iux}}{iu} - 1 \cdot \frac{e^{iux}}{i^2 u^2} \right]_{-\alpha}^{\alpha} \quad \text{by Bernoulli's rule.}$$

$$= \frac{ix}{i^2 u} [x e^{iux}]_{-\alpha}^{\alpha} - \frac{1}{i^2 u^2} [e^{iux}]_{-\alpha}^{\alpha}$$

$$= \frac{-i}{u} \{ \alpha e^{i\alpha u} - (-\alpha) e^{-i\alpha u} \} + \frac{1}{u^2} \{ e^{i\alpha u} - e^{-i\alpha u} \}$$

$$= \frac{-i}{u} \{ \alpha e^{i\alpha u} + \alpha e^{-i\alpha u} \} + \frac{1}{u^2} \{ e^{i\alpha u} - e^{-i\alpha u} \}$$

$$\text{Also, } e^{i\alpha u} = \cos \alpha u + i \sin \alpha u, \quad e^{-i\alpha u} = \cos \alpha u - i \sin \alpha u$$

$$e^{i\alpha u} + e^{-i\alpha u} = 2 \cos \alpha u, \quad e^{i\alpha u} - e^{-i\alpha u} = 2i \sin \alpha u$$

$$\text{Hence, } F(u) = \frac{-i(2\alpha \cos \alpha u)}{u} + \frac{i(2 \sin \alpha u)}{u^2}$$

$$\text{Thus, } F(u) = 2i \left[\frac{\sin \alpha u}{u^2} - \frac{\alpha \cos \alpha u}{u} \right]$$

③ If $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$

find the Fourier transform of $f(x)$ & hence find the value of

$$(i) \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$$

$$(ii) \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos\left(\frac{x}{2}\right) dx$$

$$\rightarrow F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx = \int_{-1}^1 (1-x^2) e^{iux} dx,$$

$$\therefore f(x) = 0 \text{ for } |x| \geq 1 \text{ \& } 1-x^2 \text{ for } |x| < 1$$

$$\therefore F(u) = \left[(1-x^2) \frac{e^{iux}}{iu} - (-2x) \frac{e^{iux}}{i^2 u^2} + (-2) \frac{e^{iux}}{i^3 u^3} \right]_{x=-1}^1, \text{ by Bernoulli's rule.}$$

$$= \frac{-i}{u} \left[(1-x^2) e^{iux} \right]_{x=-1}^1 - \frac{2}{u^2} \left[x e^{iux} \right]_{x=-1}^1 - \frac{2i}{u^3} \left[e^{iux} \right]_{x=-1}^1$$

$$\left[i^2 = -1, \frac{1}{i} = -i, \frac{1}{i^3} = i \right]$$

$$F(u) = \frac{-i}{u} (0-0) - \frac{2}{u^2} \{ 1 \cdot e^{iu} - (-1) e^{-iu} \} - \frac{2i}{u^3} (e^{iu} - e^{-iu})$$

$$= -\frac{2}{u^2} (e^{iu} + e^{-iu}) - \frac{2i}{u^3} (e^{iu} - e^{-iu})$$

$$\text{But, } e^{iu} = \cos u + i \sin u, \quad e^{-iu} = \cos u - i \sin u$$

$$\therefore e^{iu} + e^{-iu} = 2 \cos u, \quad e^{iu} - e^{-iu} = 2i \sin u$$

$$\therefore \text{Hence, } F(u) = \frac{-4 \cos u}{u^2} + \frac{4 \sin u}{u^3}$$

$$\text{Thus, } \boxed{F(u) = 4 \left[\frac{\sin u - u \cos u}{u^3} \right]}$$

$$\text{Let us evaluate } \int_{-\infty}^{\infty} \frac{x \cos x - \sin x}{x^3} dx$$

By inverse Fourier transform, we have,

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) e^{-iux} du \quad \text{--- (1)}$$

If $x=0$; $f(x) = 1 - 0^2 = 1$ at

By putting $x=0$ in the integral & using the expression of $f(u)$ we get,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} 4 \left[\frac{\sin u - u \cos u}{u^3} \right] e^0 du = f(0) = 1.$$

$$\int_{-\infty}^{\infty} \frac{\sin u - u \cos u}{u^3} du = \frac{2\pi}{4} = \frac{\pi}{2}$$

If u is changed to $-u$, the expression $\frac{\sin u - u \cos u}{u^3}$ becomes

$$\frac{\sin(-u) - (-u) \cos(-u)}{(-u)^3} = \frac{\sin u - u \cos u}{u^3} \text{ itself.}$$

Therefore the function is even and hence the integral from $-\infty$ to ∞ is twice the integral from 0 to ∞ .

$$\therefore 2 \int_0^{\infty} \frac{\sin u - u \cos u}{u^3} du = \frac{\pi}{2} \quad (01)$$

$$\int_0^{\infty} \frac{u \cos u - \sin u}{u^3} du = -\frac{\pi}{4}$$

changing u to x we get

$$\boxed{\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx = -\frac{\pi}{4}}$$

next, let us evaluate $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos\left(\frac{x}{2}\right) dx$

putting $x = 1/2$ in (1) we have,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} H \left[\frac{\sin u - u \cos u}{u^3} \right] e^{-\frac{iu}{2}} du = f\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

ie, $\frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin u - u \cos u}{u^3} \left[\cos \frac{u}{2} - i \sin \frac{u}{2} \right] du = \frac{3}{4}$

Equating the real parts on both sides we get,

$$\int_{-\infty}^{\infty} \frac{\sin u - u \cos u}{u^3} \cos \frac{u}{2} du = \frac{3\pi}{8}$$

ie, $2 \int_0^{\infty} \frac{\sin u - u \cos u}{u^3} \cos \frac{u}{2} du = \frac{3\pi}{8}$, since the integrand is even.

Dividing by 2, changing the sign & writing x in place of u we get,

$$\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx = -\frac{3\pi}{16}$$

(4) Find the Fourier transform of $f(x) = e^{-|x|}$

→ Fourier transform of $f(x)$ is given by

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx.$$

Here $f(x) = e^{-|x|} = \begin{cases} e^{-x} & \text{for } x \geq 0 \\ e^x & \text{for } x < 0 \end{cases}$

$$\therefore F(u) = \int_{-\infty}^0 e^x e^{iux} dx + \int_0^{\infty} e^{-x} e^{iux} dx$$

$$\begin{aligned} F(u) &= \int_{-\infty}^0 e^{(1+iu)x} dx + \int_0^{\infty} e^{-(1-iu)x} dx \\ &= \left[\frac{e^{(1+iu)x}}{1+iu} \right]_{x=-\infty}^0 + \left[\frac{e^{-(1-iu)x}}{-(1-iu)} \right]_{x=0}^{\infty} \end{aligned}$$

$$= \left[\frac{1}{1+iu} - 0 \right] + \left[0 - \frac{1}{-(1-iu)} \right]$$

$$= \frac{1}{1+iu} + \frac{1}{1-iu} = \frac{2}{1-i^2 u^2} = \frac{2}{1+u^2}$$

Thus
$$F(u) = \frac{2}{1+u^2}$$

Ex. 5 M.P. Find the Fourier transform of

$$f(x) = \begin{cases} 1-|x| & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases} \quad \text{and hence deduce that}$$

$$\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

$$\rightarrow F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{iux} dx = \int_{-1}^1 (1-|x|) e^{iux} dx$$

$$\begin{aligned} F[f(x)] &= \int_{-1}^0 [1-(-x)] e^{iux} dx + \int_0^1 [1-(+x)] e^{iux} dx \\ &= \int_{-1}^0 (1+x) e^{iux} dx + \int_0^1 (1-x) e^{iux} dx \end{aligned}$$

$$\begin{aligned}
 F[f(x)] &= \left[(1+x) \frac{e^{iux}}{iu} - (1) \frac{e^{iux}}{(iu)^2} \right]_{-1}^0 + \left[(1-x) \frac{e^{iux}}{iu} - (-1) \frac{e^{iux}}{(iu)^2} \right]_0^1 \\
 &= \frac{1}{iu} (1-0) + \frac{1}{u^2} (1 - e^{-iu}) + \frac{1}{iu} (0-1) - \frac{1}{u^2} (e^{iu} - 1) \\
 &= \frac{2}{u^2} - \frac{1}{u^2} (e^{iu} + e^{-iu}) \\
 &= \frac{2}{u^2} - \frac{2 \cos u}{u^2} = \frac{2(1 - \cos u)}{u^2}
 \end{aligned}$$

$$\text{Thus, } \boxed{F[f(x)] = \frac{4 \sin^2(u/2)}{u^2} = f(u)}$$

Now, $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du$, being the inverse Fourier transform.

$$\text{i.e., } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4 \sin^2(u/2)}{u^2} e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u/2)}{(u/2)^2} e^{-iux} du$$

putting $x=0$ we have $f(0)=1$ by the definition of $f(x)$.

$$\therefore 1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u/2)}{(u/2)^2} du$$

put $u/2 = t \therefore du = 2dt$ & t also varies from $-\infty$ to ∞

$$\text{Hence, } 1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} \cdot 2 dt = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} dt$$

$$(or) \quad 1 = \frac{1}{\pi} \cdot 2 \int_0^{\infty} \frac{\sin^2 t}{t^2} dt, \text{ since the integrand is even.}$$

$$\text{Thus, } \boxed{\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}}$$

⑥ Find the complex Fourier transform of $e^{-a^2 x^2}$, $a > 0$,
 Hence deduce that $e^{-x^2/2}$ is self reciprocal in respect
 of the complex Fourier transform.

→ $F(u) = F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{iux} dx$

i.e., $F(u) = \int_{-\infty}^{\infty} e^{-a^2 x^2} \cdot e^{iux} dx = \int_{-\infty}^{\infty} e^{-a^2 \left[x^2 - \frac{iux}{a^2} \right]} dx$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left[x^2 - 2x \cdot \frac{i u}{2a^2} + \frac{i^2 u^2}{4a^4} - \frac{i^2 u^2}{4a^4} \right]} dx$$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left[x - \frac{i u}{2a^2} \right]^2} \cdot e^{-u^2/4a^2} dx$$

put $a \left[x - \frac{i u}{2a^2} \right] = t$ $\therefore dx = \frac{dt}{a}$ &

t also varies from $-\infty$ to ∞ .

now, $F(u) = e^{-u^2/4a^2} \int_{-\infty}^{\infty} e^{-t^2} \cdot \frac{dt}{a}$ and

we know that $\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$

then $\boxed{F(u) = \frac{\sqrt{\pi}}{a} e^{-u^2/4a^2}}$

now taking, $a^2 = 1/2$ we have,

$$F(u) = F[e^{-x^2/2}] = \frac{\sqrt{\pi}}{(1/\sqrt{2})} e^{-u^2/2} = \sqrt{2\pi} e^{-u^2/2}$$

It can be seen that the Fourier transform $e^{-x^2/2}$
 is a constant times $e^{-u^2/2}$.

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the function $e^{-x^2/2}$ & $e^{-u^2/2}$ are same but for the change in the variable.

Hence we conclude that $e^{-x^2/2}$ is self reciprocal under complex Fourier transform.

(7) Find the inverse Fourier transform of e^{-u^2}
 → we have the inverse Fourier transform.

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-u^2} \cdot e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u^2 + iux)} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u^2 + 2 \cdot u \cdot \frac{i x}{2} + \frac{i^2 x^2}{4} - \frac{i^2 x^2}{4})} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u + \frac{i x}{2})^2} \cdot e^{-x^2/4} du$$

put, $u + \frac{i x}{2} = t$ ∴ $du = dt$ & t varies from $-\infty$ to ∞

$$\text{Hence, } f(x) = \frac{e^{-x^2/4}}{2\pi} \int_{-\infty}^{\infty} e^{-t^2} dt = \frac{e^{-x^2/4}}{2\pi} \sqrt{\pi} = \frac{e^{-x^2/4}}{2\sqrt{\pi}}$$

Thus the required inverse Fourier transform is

$$\frac{e^{-x^2/4}}{2\sqrt{\pi}}$$

* Fourier Cosine and Fourier Sine Transforms

Inverse Fourier cosine & Inverse Fourier sine transform.

If $f(x)$ is defined for all positive values of x , we define the following.

$$F_c[f(x)] = \int_0^{\infty} f(x) \cos ux \, dx = F_c(u) \text{ -----}$$

Fourier cosine transform

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(u) \cos ux \, du \text{ --- Inverse Fourier cosine transform}$$

$$F_s[f(x)] = \int_0^{\infty} f(x) \sin ux \, dx = F_s(u) \text{ --- Fourier sine transform}$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(u) \sin ux \, du \text{ --- Inverse Fourier sine transform.}$$

* Note :- The following properties concerning Fourier cosine and Fourier sine transform can easily be established as in the case of Fourier transforms.

* Properties

① Linearity Property

$$\begin{aligned} F_c[C_1 f_1(x) + C_2 f_2(x) + \dots + C_n f_n(x)] \\ = C_1 F_c[f_1(x)] + C_2 F_c[f_2(x)] + \dots + C_n F_c[f_n(x)] \end{aligned}$$

(2) change of scale property

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$$\text{If } F_c[f(x)] = F_c(u) \text{ then } F_c[f(ax)] = \frac{1}{a} F_c\left[\frac{u}{a}\right]$$

These two properties continue to hold good in the case of Fourier sine transform also.

(3) Modulation properties

$$\text{If } F_s[f(x)] = F_s(u) \text{ \& } F_c[f(x)] = F_c(u) \text{ then}$$

$$\textcircled{i} \quad \underline{F_s[f(x) \cos ax] = \frac{1}{2} [F_s(u+a) + F_s(u-a)]}$$

$$\begin{aligned} \rightarrow F_s[f(x) \cos ax] &= \int_0^\infty f(x) \cos ax \cdot \sin ux \, dx \\ &= \int_0^\infty f(x) \cdot \frac{1}{2} [\sin(u+a)x + \sin(u-a)x] \, dx \\ &= \frac{1}{2} \left[\int_0^\infty f(x) \sin(u+a)x \, dx + \int_0^\infty f(x) \sin(u-a)x \, dx \right] \end{aligned}$$

$$F_s[f(x) \cos ax] = \frac{1}{2} [F_s(u+a) + F_s(u-a)]$$

$$\textcircled{ii} \quad F_s[f(x) \sin ax] = \int_0^\infty f(x) \sin ax \cdot \sin ux \, dx.$$

$$\begin{aligned} \rightarrow F_s[f(x) \sin ax] &= \int_0^\infty f(x) \sin ax \cdot \sin ux \, dx \\ &= \int_0^\infty f(x) \cdot \frac{1}{2} [\cos(u-a)x - \cos(u+a)x] \, dx \\ &= \frac{1}{2} \left[\int_0^\infty f(x) \cos(u-a)x \, dx - \int_0^\infty f(x) \cos(u+a)x \, dx \right] \end{aligned}$$

$$F_s[f(x) \sin ax] = \frac{1}{2} [F_c(u-a) - F_c(u+a)]$$

(iii)

(17)

$$F_c[f(x) \cos ax] = \frac{1}{2} [F_c(u+a) + F_c(u-a)]$$

$$\begin{aligned} \rightarrow F_c[f(x) \cos ax] &= \int_0^\infty f(x) \cos ax \cdot \cos ux \, dx \\ &= \int_0^\infty f(x) \cdot \frac{1}{2} [\cos(u+a)x + \cos(u-a)x] \, dx \\ &= \frac{1}{2} \left[\int_0^\infty f(x) \cos(u+a)x \, dx + \int_0^\infty f(x) \cos(u-a)x \, dx \right] \end{aligned}$$

$$F_c[f(x) \cos ax] = \frac{1}{2} [F_c(u+a) + F_c(u-a)]$$

(iv)

$$F_c[f(x) \sin ax] = \frac{1}{2} [F_s(u+a) - F_s(u-a)]$$

$$\begin{aligned} \rightarrow F_c[f(x) \sin ax] &= \int_0^\infty f(x) \sin ax \cdot \cos ux \, dx \\ &= \int_0^\infty f(x) \cdot \frac{1}{2} [\sin(a+u)x + \sin(a-u)x] \, dx \\ &= \frac{1}{2} \left[\int_0^\infty f(x) \sin(u+a)x \, dx - \int_0^\infty f(x) \sin(u-a)x \, dx \right] \end{aligned}$$

$$F_c[f(x) \sin ax] = \frac{1}{2} [F_s(u+a) - F_s(u-a)]$$

PROBLEMS

M.P

(1) Find the Fourier sine and cosine transforms of $f(x) = e^{-\lambda x}$, $x > 0$.

→ Fourier sine and cosine transforms are given by

$$F_s(u) = \int_0^{\infty} f(x) \sin ux \, dx \quad \&$$

$$F_c(u) = \int_0^{\infty} f(x) \cos ux \, dx$$

$$\therefore F_s(u) = \int_0^{\infty} e^{-\lambda x} \sin ux \, dx$$

$$F_s(u) = \left[\frac{e^{-\lambda x}}{(-\lambda)^2 + u^2} (-\lambda \sin ux - u \cos ux) \right]_0^{\infty}$$

by using the standard formula,

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax} (a \sin bx - b \cos bx)}{a^2 + b^2}.$$

But, $e^{-\lambda x} \rightarrow 0$ as $x \rightarrow \infty$, $e^0 = 1$, $\cos 0 = 1$, & $\sin 0 = 0$.

$$\text{Thus, } \boxed{F_s(u) = \frac{u}{\lambda^2 + u^2}}$$

$$\text{Also, } F_c(u) = \int_0^{\infty} f(x) \cos ux \, dx = \int_0^{\infty} e^{-\lambda x} \cos ux \, dx$$

$$= \left[\frac{e^{-\lambda x}}{(-\lambda)^2 + u^2} (-\lambda \cos ux + u \sin ux) \right]_{x=0}^{\infty}$$

$$= \frac{\lambda}{\lambda^2 + u^2}$$

by using $\int e^{ax} \cos bx \, dx = \frac{e^{ax} (a \cos bx + b \sin bx)}{a^2 + b^2}$

Thus, $\boxed{F_c(u) = \frac{x}{x^2 + u^2}}$ //

(2) obtain the Fourier cosine transform of the function.

$$f(x) = \begin{cases} 4x, & 0 < x < 1 \\ 4-x, & 1 < x < 4 \\ 0, & x > 4 \end{cases}$$

→ Fourier cosine transform is given by

$$F_c(u) = \int_0^{\infty} f(x) \cos ux \, dx$$

$$= \int_0^1 f(x) \cos ux \, dx + \int_1^4 f(x) \cos ux \, dx + \int_4^{\infty} f(x) \cos ux \, dx$$

$$\therefore F_c(u) = \int_0^1 4x \cos ux \, dx + \int_1^4 (4-x) \cos ux \, dx + \int_4^{\infty} 0 \cdot \cos ux \, dx$$

Applying Bernoulli's rule to integrals we have,

$$F_c(u) = \left[4x \cdot \frac{\sin ux}{u} - 4 \frac{-\cos ux}{u^2} \right]_0^1 + \left[(4-x) \frac{\sin ux}{u} - (-1) \frac{-\cos ux}{u^2} \right]_1^4$$

+ 0

$$= \frac{4}{u} [x \sin ux]_0^1 + \frac{4}{u^2} [\cos ux]_0^1 + \frac{1}{u} [(4-x) \sin ux]_1^4 - \frac{1}{u^2} [\cos ux]_1^4$$

$$= \frac{4}{u} (\sin u - 0) + \frac{4}{u^2} (\cos u - 1) + \frac{1}{u} (0 - 3 \sin u) - \frac{1}{u^2} (\cos 4u - \cos u)$$

(20)

$$F_c(u) = \frac{4}{u} \sin u + \frac{4}{u^2} \cos u - \frac{4}{u^2} - \frac{3}{u} \sin u - \frac{1}{u^2} \cos 4u + \frac{1}{u^2} \cos u$$

gives,
$$F_c(u) = \frac{1}{u} \sin u + \frac{5 \cos u - 4}{u^2} - \frac{1}{u^2} \cos 4u$$

* Similar problem

③ Find the Fourier cosine transform of $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2-x & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$

(4) Find the Fourier sine transform of $f(x) = e^{-|x|}$ (21)
 here evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx, m > 0.$

→ Fourier sine transform is given by

$$F_s(u) = \int_0^{\infty} f(x) \sin ux dx$$

$$F_s(u) = \int_0^{\infty} e^{-|x|} \sin ux dx = \int_0^{\infty} e^{-x} \sin ux dx,$$

since $|x| = x, x > 0.$

$$e^{-|x|} = \begin{cases} e^{-x} & \text{for } x \geq 0 \\ e^x & \text{for } x < 0 \end{cases}$$

$$F_s(u) = \left[\frac{e^{-x}(-1 \sin ux - u \cos ux)}{(-1)^2 + u^2} \right]_0^{\infty}$$

But, $e^{-x} \rightarrow 0$ as $x \rightarrow \infty$, $e^0 = 1$, $\cos 0 = 1$, $\sin 0 = 0$

$$\text{Thus, } F_s(u) = \frac{u}{1+u^2}$$

By inverse Fourier sine transform we have,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(u) \sin ux du.$$

$$\frac{\pi}{2} f(x) = \int_0^{\infty} \frac{u}{1+u^2} \sin ux du$$

putting $x=m$ where $m > 0$ we have $f(x) = e^{-|m|} = e^{-m}$

$$\therefore \int_0^{\infty} \frac{u \sin mu}{1+u^2} du = \frac{\pi}{2} e^{-m}$$

Thus by changing the variable u to x ,

$$\boxed{\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}}$$

(5) Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$

→ we have $F_s(u) = \int_0^\infty f(x) \sin ux \, dx$ &

$$\text{let } f(x) = \frac{e^{-ax}}{x}$$

$$\text{i.e., } F_s(u) = \int_0^\infty \frac{e^{-ax}}{x} \sin ux \, dx. \quad \text{--- (1)}$$

we cannot evaluate this integral directly & hence we proceed as follows.

$$\therefore \frac{d}{du} [F_s(u)] = \int_0^\infty \frac{e^{-ax}}{x} \frac{\partial}{\partial u} (\sin ux) \, dx.$$

$$= \int_0^\infty \frac{e^{-ax}}{x} x \cos ux \, dx$$

$$\frac{d}{du} [F_s(u)] = \int_0^\infty e^{-ax} \cos ux \, dx$$

$$= \left[\frac{e^{-ax}}{a^2 + u^2} (-a \cos ux + u \sin ux) \right]_{x=0}^\infty$$

$$= \frac{1}{a^2 + u^2} (0 + a) = \frac{a}{a^2 + u^2}$$

hence, $\frac{d}{du} [F_s(u)] = \frac{a}{a^2 + u^2}$ & by integrating w.r.t u we get,

$$F_s(u) = \tan^{-1}(u/a) + c$$

to evaluate c , let us put $u = 0$

$$\therefore F_s(0) = \tan^{-1}(0) + c$$

But, $F_s(0) = 0$ from (1) & hence $c = 0$.

$$\text{Thus, } \boxed{F_s(u) = \tan^{-1}(u/a)}$$

⑥ Find the Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$ (23)

→ By the definition, $F_c[f(x)] = \int_0^\infty f(x) \cos ux \, dx = F_c(u)$

$$\text{i.e., } F_c(u) = \int_0^\infty \frac{1}{1+x^2} \cos ux \, dx \quad \text{--- (1)}$$

diff under integral sign.

we cannot evaluate the RHS directly & hence we proceed as follows.

diff w.r.t u

$$\frac{dF_c(u)}{du} = F'_c(u) = \int_0^\infty \frac{1}{1+x^2} (-\sin ux) x \, dx$$

$$\text{i.e., } F'_c(u) = - \int_0^\infty \frac{x}{1+x^2} \sin ux \, dx$$

we can't integrate directly.

(we cannot evaluate RHS now also and hence modify the integrand).

$$F'_c(u) = - \int_0^\infty \frac{x^2}{x(1+x^2)} \sin ux \, dx$$

$$= - \int_0^\infty \frac{(1+x^2) - 1}{x(1+x^2)} \sin ux \, dx$$

$$F'_c(u) = - \int_0^\infty \frac{\sin ux}{x} \, dx + \int_0^\infty \frac{\sin ux}{x(1+x^2)} \, dx.$$

we note that $\int_0^\infty \frac{\sin ux}{x} \, dx = \frac{\pi}{2}$ & hence we have

$$F'_c(u) = -\frac{\pi}{2} + \int_0^\infty \frac{\sin ux}{x(1+x^2)} \, dx \quad \text{--- (2)}$$

Diff again w.r.t u we have,

$$F''_c(u) = \int_0^\infty \frac{\cos ux \cdot x}{x(1+x^2)} \, dx = \int_0^\infty \frac{\cos ux}{1+x^2} \, dx$$

$\therefore F_c''(u) = F_c(u)$, by using ①.

$$(ii) F_c''(u) - F_c(u) = 0 \quad \&$$

this is 2nd order ODE of the form

$$(D^2 - 1)F_c(u) = 0, \quad \text{where } D = \frac{d}{du}$$

$$\text{A.E. is } m^2 - 1 = 0$$

$$m = 1, -1$$

The general solⁿ is given by

$$F_c(u) = c_1 e^u + c_2 e^{-u} \quad \text{--- ③}$$

$$\text{But from ①, } F_c(0) = \int_0^\infty \frac{1}{1+x^2} dx$$

$$F_c(0) = \left[\tan^{-1} x \right]_0^\infty = \pi/2$$

$$\text{So we have, } c_1 + c_2 = \pi/2 \quad \text{--- ④}$$

$$\text{Also from ③, } F_c'(u) = c_1 e^u - c_2 e^{-u}$$

$$\text{hence } F_c'(0) = c_1 - c_2$$

$$\text{From ② } F_c'(0) = -\pi/2 + 0 = -\pi/2$$

$$\text{Hence, we have, } c_1 - c_2 = -\pi/2 \quad \text{--- ⑤}$$

$$\text{By solving ④ \& ⑤ we get, } c_1 = 0 \quad \& \quad c_2 = \pi/2$$

$$\text{Using these values in ③ we have } F_c(u) = (\pi/2) e^{-u}$$

$$\text{Thus the required, } \boxed{F_c\left[\frac{1}{1+x^2}\right] = \frac{\pi}{2} e^{-u}} \quad //$$

⑦ Find the Fourier sine transform of $f(x) = \frac{1}{x(1+x^2)}$ (25)

$$\rightarrow F_s[f(x)] = \int_0^\infty f(x) \sin ux \, dx = F_s(u)$$

$$\text{i.e. } F_s(u) = \int_0^\infty \frac{\sin ux}{x(1+x^2)} \, dx \quad \text{--- (1)}$$

we cannot evaluate the integral directly, hence we proceed as follows.

$$\therefore F'_s(u) = \int_0^\infty \frac{x \cos ux}{x(1+x^2)} \, dx = \int_0^\infty \frac{\cos ux}{(1+x^2)} \, dx \quad \text{--- (2)}$$

Diff again w.r.t u ,

$$F''_s(u) = \int_0^\infty \frac{-x \sin ux}{(1+x^2)} \, dx = - \int_0^\infty \frac{x^2 \sin ux}{x(1+x^2)} \, dx$$

$$\text{i.e. } F''_s(u) = - \int_0^\infty \frac{(1+x^2)-1}{x(1+x^2)} \sin ux \, dx$$

$$F''_s(u) = - \int_0^\infty \underbrace{\frac{\sin ux}{x}}_{\frac{\pi}{2}} \, dx + \int_0^\infty \frac{\sin ux}{x(1+x^2)} \, dx$$

$$\text{i.e. } F''_s(u) = -\frac{\pi}{2} + F_s(u)$$

$$F''_s(u) - F_s(u) = -\frac{\pi}{2}$$

$$\text{i.e. } [D^2 - 1] F_s(u) = -\frac{\pi}{2} \quad \text{where } D = \frac{d}{du}$$

$$\text{A.E. is } m^2 - 1 = 0$$

$$\therefore m = \pm 1$$

C.F. is given by $C_1 e^u + C_2 e^{-u}$

$$\text{Also, P.I.} = \frac{-\pi/2}{D^2 - 1} = \frac{-\pi/2}{-1} = \frac{\pi}{2}$$

The general solution is given by $CF + PI$

i.e., $F_s(u) = c_1 e^u + c_2 e^{-u} + \pi/2$ — (3)

we need to find c_1 & c_2 .

$$F_s'(u) = c_1 e^u - c_2 e^{-u} \quad \text{--- (4)}$$

putting $u=0$ in (3) & (4) we have

$$F_s(0) = c_1 + c_2 + \pi/2 \quad \&$$

$$F_s'(0) = c_1 - c_2$$

But $F_s(0) = 0$ from (1) &

$$F_s'(0) = \int_0^\infty \frac{1}{1+x^2} dx \quad \text{from (2)}$$

i.e., $F_s'(0) = [\tan^{-1} x]_0^\infty = \pi/2$

we have the system of equations

$$c_1 + c_2 + \pi/2 = 0 \quad \& \quad c_1 - c_2 = \pi/2$$

By solving we get $c_1 = 0$,
 $c_2 = -\pi/2$ &

we substitute these values in (3).

$$\text{Thus, } \boxed{F_s(u) = \frac{\pi}{2} (1 - e^{-u})}$$

⑧ Find the inverse Fourier transform of e^{-u^2}
 → we have the inverse Fourier transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-u^2} \cdot e^{-iux} du$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u^2 + iux)} du$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u^2 + 2 \cdot u \cdot \frac{ix}{2} + \frac{i^2 x^2}{4} - \frac{i^2 x^2}{4})} du$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(u + \frac{ix}{2})^2} \cdot e^{-x^2/4} du$$

put $u + \frac{ix}{2} = t$

$\therefore du = dt$ & t varies from $-\infty$ to ∞ ,

Hence, $f(x) = \frac{e^{-x^2/4}}{2\pi} \int_{-\infty}^{\infty} e^{-t^2} dt$

$$= \frac{e^{-x^2/4}}{2\pi} \sqrt{\pi}$$

$$f(x) = \frac{e^{-x^2/4}}{2\sqrt{\pi}}$$

Thus the required inverse Fourier transform

is $\boxed{\frac{e^{-x^2/4}}{2\sqrt{\pi}}}$ ✓

(9) Find the inverse Fourier sine transform of

$$f_s(x) = \frac{1}{x} e^{-ax}, \quad a > 0$$

→ By data, $f_s(x) = \frac{e^{-ax}}{x}$

$$\begin{aligned} \text{i.e. } F_s[f(x)] &= \int_0^\infty f(x) \sin ax \, dx \\ &= f_s(x) \quad \text{hence} \end{aligned}$$

$$f(x) = \frac{2}{\pi} \int_0^\infty f_s(x) \sin ax \, dx,$$

being the inverse Fourier sine transform.

$$\begin{aligned} \text{i.e. } f(x) &= \frac{2}{\pi} \int_0^\infty \frac{e^{-ax}}{x} \sin ax \, dx \\ &= \frac{2}{\pi} \phi(x) \quad (\text{say}) \end{aligned}$$

$$\text{where } \phi(x) = \int_0^\infty \frac{e^{-ax}}{x} \sin ax \, dx$$

we cannot evaluate the integral directly & hence proceed as follows.

$$\phi'(x) = \int_0^\infty \frac{e^{-ax}}{x} \cos ax \cdot x \, dx$$

$$= \int_0^\infty e^{-ax} \cos ax \, dx$$

$$\phi'(x) = \left[\frac{e^{-ax}}{a^2 + x^2} (-a \cos ax + x \sin ax) \right]_{x=0}^\infty, \quad \text{by a standard formula.}$$

$$\phi'(x) = \frac{1}{a^2+x^2} \{0 - (-a)\}$$

$$= \frac{a}{a^2+x^2}$$

$$\therefore \phi(x) = \int \frac{a}{a^2+x^2} dx + C$$

$$\phi(x) = \tan^{-1}(x/a) + C$$

To evaluate C let us put $x=0$.

$$\phi(0) = \tan^{-1}(0) + C \quad (1)$$

$$0 = 0 + C$$

$$\therefore \boxed{C=0}$$

$$\text{Hence } \phi(0) = 0.$$

$$\text{Hence } \phi(x) = \tan^{-1}(x/a) \text{ \& } \phi$$

$$\text{we have } f(x) = \frac{2}{\pi} \phi(x)$$

$$\text{Hence, } \boxed{f(x) = \frac{2}{\pi} \tan^{-1}(x/a)} //$$

(10) Find the Fourier cosine transform of e^{-ax} & hence deduce the Fourier cosine transform of $x e^{-ax}$. Further evaluate $\int_0^{\infty} \frac{\cos x}{x^2+a^2} dx$

$$\rightarrow \text{we have } \mathcal{F}_c[e^{-ax}] = \frac{a}{a^2+u^2}$$

$$\text{i.e., } \int_0^{\infty} e^{-ax} \cos ux \, dx = \frac{a}{a^2+u^2} \quad \text{--- (1)}$$

Def 1 ① w.r.t a on both sides we get.

$$\int_0^{\infty} e^{-ax} (-x) \cos ux \, dx = \frac{(a^2 + u^2)(1) - 2a^2}{(a^2 + u^2)^2}$$

$$= \frac{u^2 - a^2}{(a^2 + u^2)^2}$$

$$(ii) \int_0^{\infty} (x e^{-ax}) \cos ux \, dx = \frac{a^2 - u^2}{(a^2 + u^2)^2}$$

$$\text{Thus in, } F_c[xe^{-ax}] = \frac{a^2 - u^2}{a^2 + u^2}$$

$$\text{Further, } F_c[e^{-ax}] = \frac{a}{a^2 + u^2}$$

$$\Rightarrow e^{-ax} = \frac{2}{\pi} \int_0^{\infty} \frac{a}{a^2 + u^2} \cos ux \, du,$$

by inverse cosine transform.

$$(iii) \int_0^{\infty} \frac{\cos ux}{a^2 + u^2} \, du = \frac{\pi}{2a} e^{-ax}$$

let us change, u to λ and u to x .

$$\text{Thus, } \boxed{\int_0^{\infty} \frac{\cos \lambda x}{a^2 + x^2} \, dx = \frac{\pi}{2a} e^{-ax}} //$$

⑪ solve the integral equation : $\int_0^{\infty} f(x) \cos \lambda x \, dx = e^{-ax}$

→ we have by inverse Fourier cosine transform,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F(\lambda) \cos \lambda x \, d\lambda \quad \&$$

(31)

$$F(x) = e^{-ax} \text{ by data.}$$

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^{\infty} e^{-ax} \cos \alpha x \, d\alpha \\ &= \frac{2}{\pi} \left[\frac{e^{-ax}}{(-a)^2 + x^2} (-a \cos \alpha x + \pi \sin \alpha x) \right]_{\alpha=0}^{\infty} \end{aligned}$$

Thus,
$$f(x) = \frac{2a}{\pi(a^2 + x^2)}$$

(12) solve the integral equation

$$\int_0^{\infty} f(t) \cos \alpha t \, dt = \begin{cases} 1-x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases} \text{ \& hence evaluate}$$

$$\int_0^{\infty} \frac{\sin^2 t}{t^2} \, dt$$

→ Here we have to find $f(t)$ & we shall consider the inverse Fourier cosine transform with.

$$F(x) = \begin{cases} 1-x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F(x) \cos \alpha t \, d\alpha$$

$$f(t) = \frac{2}{\pi} \int_0^1 (1-x) \cos \alpha t \, d\alpha, \quad \text{since } F(x) = 0 \text{ for } x > 1$$

$$= \frac{2}{\pi} \left[(1-x) \frac{\sin \alpha t}{t} - (-1) \left[\frac{-\cos \alpha t}{t^2} \right] \right]_{\alpha=0}^1$$

$$= \frac{2}{\pi t} \left[(1-x) \sin \alpha t \right]_{\alpha=0}^1 - \frac{2}{\pi t^2} \left[\cos \alpha t \right]_{\alpha=0}^1$$

$$f(\theta) = \frac{2}{\pi\theta} [0-0] - \frac{2}{\pi\theta^2} [\cos\theta - 1]$$

$$\text{i.e. } f(\theta) = \frac{2(1-\cos\theta)}{\pi\theta^2} = \frac{2 \cdot 2\sin^2(\theta/2)}{\pi\theta^2}$$

$$= \frac{4\sin^2(\theta/2)}{\pi\theta^2}$$

Hence we now have $\int_0^\infty \frac{4\sin^2(\theta/2)}{\pi\theta^2} \cos\alpha\theta d\theta = F(\alpha)$

i.e., $\int_0^\infty \frac{\sin^2(\theta/2)}{(\theta/2)^2} \cos\alpha\theta d\theta = \pi F(\alpha)$

putting $\theta/2 = t$,

$d\theta = 2dt$ & t varies from 0 to ∞ .

$\therefore \int_0^\infty \frac{\sin^2 t}{t^2} \cos(2\alpha t) \cdot 2dt = \pi F(\alpha)$

i.e., $\int_0^\infty \frac{\sin^2 t}{t^2} \cos(2\alpha t) dt = \frac{\pi}{2} F(\alpha)$

putting $\alpha = 0$, $F(\alpha) = 1 - 0 = 1$

Hence, $\boxed{\int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}}$

(13) solve the following integral equation:

$$\int_0^\infty f(x) \sin \alpha x dx = \begin{cases} 10, & 0 \leq \alpha < 1 \\ 20, & 1 \leq \alpha < 2 \\ 0, & \alpha \geq 2 \end{cases}$$

→

$$\rightarrow \text{If } \int_0^{\infty} f(x) \sin \alpha x \, dx = F_s(\alpha)$$

then we have to find $f(x)$ where,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(\alpha) \sin \alpha x \, d\alpha$$

$$f(x) = \frac{2}{\pi} \left[\int_0^1 10 \sin \alpha x \, d\alpha + \int_1^2 20 \sin \alpha x \, d\alpha + \int_2^{\infty} 0 \cdot \sin \alpha x \, d\alpha \right]$$

$$= \frac{2}{\pi} \left\{ \left[-\frac{10 \cos \alpha x}{x} \right]_0^1 + \left[-\frac{20 \cos \alpha x}{x} \right]_1^2 \right\}$$

$$= -\frac{20}{\pi x} \left\{ [\cos \alpha x]_0^1 + 2 [\cos \alpha x]_1^2 \right\}$$

$$= -\frac{20}{\pi x} (\cos x - 1 + 2 \cos 2x - 2 \cos x)$$

$$= -\frac{20}{\pi x} (-1 - \cos x + 2 \cos 2x)$$

$$\text{Thus, } \boxed{f(x) = \frac{20}{\pi x} (1 + \cos x - 2 \cos 2x)}$$